

Version 5 analysis note instructions

I now have many more studies that I have conducted. Everything must be merged and curated into a version 5.0.0 analysis note that I will be using to request 100% unblinding.

I believe the last full analysis note written was v4.9.7 and then the prose that was written for the harvard writing sample located at /Users/emrypeets/Desktop/gp_mods/hps-gpr-emrys-validation_workflow/study_results/harvard_writing_sample_final_combinations_20260902/pdf/HPS_GPR_Harvard_Writing_Sample_Final_Combinations.pdf would be good to include. The harvard application writing could be good for the included sections, but will have to be expanded to go into more detail of the following many studies.

Agent Guidelines:

Be token efficient when possible, make a writing managing agent to delegate tasks appropriately and dont do too many parallel tasks, if toys are requested only do that if you dont time out, otherwise include placeholders for now. Ensure there is clear writing throughout the document and that it reads well. Pause progress at 3:10 PM today and make a handoff document with instructions at that point.

The change log should be the first section of the appendix and summarize each of the recent studies and the documents that were made and how they influenced this final "unblinding request version 5 document" The appendix should be clean and have its own table of contents after the change log, the studies should be less disorganized and explain when historic studies were conducted and what the significant changes made.

In "HPS_GPR_Analysis_Note_v4p9p12_targeted_tail_refinement.pdf"

Take the figures

1 (very important figure, underneath I would like to see the asymptotic pvalues as an additon), 2, 3, 4(remove "(optimized support)" from 2021 title and then remove this line from caption: "native-10% configuration with "), 5, 6 (the extraction figures in this pdf are messed up I believe) — I would like figures 5-6 to be continued where the statistical tails do not resolve the pvalues, 15, 16 [15, 16 do a good little study on varying the mass resolution and shows the impact it would have on the upper limit and significances, so this could be added into the results section as (varying mass resolution),

- figure 16 should remove this line from caption " Width and mass selection are post-selection diagnostics, not independent significance gains or a measurement of resolution miscalibration. Observed data; zero new toys",].
- . Some version of figure 14 should already exist at the start of the mass resolution portion of the note so add it if it doesn't. Also in the main mass resolution portion, reference the studies conducted where we vary it that are later shown in the results.
- I would like the observed upper limits in coupling overlaid on each other (figure 1 observed combined limit and each of the three individual limits only)

In HPS_GPR_Analysis_Note_v4p9p13_calibrated_backgrounds.pdf

- Include figure 9 from this study, In the methodology sections of the main analysis note, we need to discuss why a profiled background is important. The strength of the relative injection that is done in figure 9 would be good to note in the caption and explanatory sentence in that section. An important thing to mention in this section is that plot shows a huge bias in injected signal for the fixed background case and a smaller and less than 1 sigma bias at 71 MeV when injected at 71 MeV, thus selecting the profiled method makes sense for 2021. note that another profiled background choice study was conducted and include figure 1 in this methodology section as well. Mention that previous studies have also shown much higher failure of exclusion power , then cite the evidence for this in the section 8 inclusion as discussed a bit further below
- From this study, calibration is investigated as a meaningful alternative approach to background modeling that impacts the upper limits more at 2016 really than any other dataset, regardless ... "Alternative background modeling through calibration"
- The entire section 8 of calibrated background model would be a nice new section in the appendix, but at the beginning of this appendix section would like a flow chart explain what the two different calibration approaches are and how they are performed for the profiled approach and the fixed approach. The "stress input" information is somewhat confusing to me. The independent validation of 90% CLs exclusion plot is a bit confusing to me as well so needs more layman descriptions. In the main methodology section it should be discussed that calibration was studies in more detail and cite the new appendix section, but ultimately there is confidence in the asymptotic methodology given the various validation studies showing limited overall bias' and so that is the retained

primary analysis approach.

- Remove the various figure inlaid text boxes that aren't legends and the information should be put in the caption instead
 - Also show what the interpolation offset method might imply and where that could be useful

Some of the calibration section should come from HPS_GPR_v4p9p14_Calibration_Explained_and_Global_Study.pdf as it is important too, but avoid being too wordy or repetitive

From HPS_GPR_v4p9p14_Calibration_Explained_and_Global_Study.pdf

- In the analysis note where global significances are selected I would like a global significance subsection in addition to the current LEE approach to include the GP global, the benefits of doing this approach should be described in this section (I think some of this should come from "Exact definition of the new GP pilot") and the full details should describe significance approach and to succinctly go into what this looks like, then in the results section I would like a series of pvalue plots for each dataset that are placed into a section, local pvalues and GP global significance studies as shown in figure 4 and summarized in this text "The separate 2015 global-significance section applies the [Ananiev-Read method](#), starting with ten complete scans, then 1,000 independent scans and 200,000 GP fields. It explains the conditional results, unresolved rare tails" we don't necessarily need the fixed background comparison as we have decided on the profiled, but this approach to doing global is interesting and I think might be good to reference. Maybe putting in placeholders for this global significance approach is important for 2016 and 2021 first due to the results shown in HPS_GPR_v4p9p15_Global_Study_2016_Full_2021_10pct.pdf maybe the stress backgrounds are just wonky? I still don't know that they are sufficient..

HPS_GPR_Analysis_Note_v4p9p16_Probability_Audit_and_Echoes

- The 2015 15-20 MeV study can be included in the appendix at the end (searching 15-20 MeV)

In:

HPS_GPR_Analysis_Note_v4p9p16_Signal_Extractions.pdf

- Add figure 14, I don't like ""signed root" since it is functionally a mapping

of significance, so I would prefer that as a label on the plot and then in the caption explain what that means. The correlation plot is nice in showing that a real signal can indeed cause various fluctuations in nearby regions, so I want the caption to reflect this. Figures 5 and 6 from HPS_GPR_v4p9p15_Global_Study_2016_Full_2021_10pct.pdf show some of the plots as well

- Find correlation plots for all the datasets and include versions where it shows “significance of induced echo oscillation” based on

In:

HPS_GPR_Analysis_Note_v4p9p16_Candidate_Removal_and_Traditional_Searches.pdf

- Figure 4 is a nice addition to the aforementioned correlation plots as it showed visibly what could happen
- A version of figure 18 should exist at the end of the results section, but delta D is a bit confusing,
- Figures 11-17 (a lot of these should go into extraction displays from data part of the results section). I like the various extraction figures of the top candidates but I would prefer smoother regions (maybe adding more binning in the display and then having straight lines through it) and showing a little bit of the sideband in each of the residual displays so it can show the gp mean and then how the signal shape is fit in bkg+signal model
 - In these, add this into the caption if its in the light grey text box, these generally should be a different color as its a bit distracting. “Bars: counting error only. Gray shading: zero-centered GP constraint width; not fitted-background error or total residual uncertainty”
 - In all of this plots and in general “signed root” is not a great metric to label these plots. I prefer local significance..
- The traditional fit studies can go into a new section of the appendix
- The joint scan of deficits can go into an appendix, but the figure 13 deficit plot should go into the extraction display result section